VegPro Smart Farm — Technical Handover Documentation

Product: VegPro Smart Farm (Scouting)

Live URL: https://scouting.kongoniriverfarm.com/

Production host: FDB (`192.168.16.14`)

Audience: VegPro IT / Admin (Dev) / implementation partner

Version: October 2026 handover

This document describes how the live system is hosted, how it is configured, and how to operate it after handover.

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1. System summary

Layer	Technology
Public website	HTTPS on IIS (ports 80/443)
Application	Next.js 15 (Node.js) on `127.0.0.1:3000`
Reverse proxy	IIS URL Rewrite + Application Request Routing (ARR)
Database	Microsoft SQL Server — database **`Scouting`** on FDB
Authentication	App users in `dbo.users` (email + bcrypt password). SQL access uses **Windows auth** as `FDB\Scouting`
Evidence / photos	Files on FDB disk (`EVIDENCE_STORAGE_PATH`)
DNS name	`scouting.kongoniriverfarm.com`
Certificate	Wildcard `*.kongoniriverfarm.com` (bound in IIS)

Users only open **https://scouting.kongoniriverfarm.com/**. They never use port 3000.

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2. Roles and seed accounts

App role	Typical user	Default seed email	Default password
`admin`	System admin	`admin@vegpro.com`	`VegPro2026!`
`supervisor` (UI: Farm manager)	Farm manager	`manager@vegpro.com`	`VegPro2026!`
`worker`	Field scout	`worker@vegpro.com`	`VegPro2026!`

Change these passwords after training.

New field workers and farm managers are created in the app: **Admin → Staff**.

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3. Server layout on FDB

Item	Path / value
App folder	`C:\Users\Scouting\Veg-pro-Smart-Farms-master`
IIS site name	`VegPro`
IIS physical path	`C:\Users\Scouting\Veg-pro-Smart-Farms-master\iis` (web.config reverse proxy)
Node listen	`0.0.0.0:3000`
Auto-start task	Scheduled Task **`VegPro App`** (runs as `FDB\Scouting` at startup)
App start script	`C:\ProgramData\VegPro\start-vegpro.cmd`
App log	`C:\ProgramData\VegPro\vegpro-app.log`
Evidence folder	`C:\Users\Scouting\Veg-pro-Smart-Farms-master\data\evidence`

3.1 How a request flows

```
Browser (HTTPS)
  → scouting.kongoniriverfarm.com:443 (IIS + certificate)
  → reverse proxy to http://127.0.0.1:3000
  → Next.js app
  → MS SQL database Scouting (Windows auth as FDB\Scouting)
```

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4. Environment configuration (`.env.local`)

File: `C:\Users\Scouting\Veg-pro-Smart-Farms-master\.env.local`

Required production keys (values on server — do not commit secrets to git):

```
DATA_BACKEND=mssql
NEXT_PUBLIC_DATA_BACKEND=mssql
MSSQL_SERVER=localhost
MSSQL_DATABASE=Scouting
MSSQL_AUTH=windows
MSSQL_TRUST_SERVER_CERTIFICATE=true
SESSION_SECRET=<long-secret>
SESSION_COOKIE_SECURE=true
EVIDENCE_STORAGE_PATH=C:\Users\Scouting\Veg-pro-Smart-Farms-master\data\evidence
```

Notes

- On FDB, `MSSQL_SERVER=localhost` is required for reliable Windows Integrated auth. Using the LAN IP can fail with "untrusted domain".
- The Node process **must** run as `FDB\Scouting` (scheduled task already does this).
- Do not enable Supabase env vars on this production host.

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5. Database

- Server: FDB SQL Server
- Database: **`Scouting`**
- Schema scripts (repo): `sql/mssql/001_schema.sql` through `007_supervisor_greenhouse_access.sql` (run in numeric order on **`Scouting`**)
- Core tables include: `users`, `farms`, `greenhouses`, `crop_categories`, `crop_varieties`, scouting rounds/records, activities, evidence-related tables as applied

5.1 Connectivity test (on FDB as Scouting)

Run these commands **on FDB**, signed in as **`FDB\Scouting`** (same account as the **VegPro App** scheduled task). Do **not** run this test from a random admin desktop unless that PC is on the farm network and you intentionally use SQL login (production uses Windows auth on the server).

```
cd /d C:\Users\Scouting\Veg-pro-Smart-Farms-master
whoami
npm run test:mssql
```

Expected success

- `whoami` → `fdb\scouting` (domain may show as `FDB\Scouting`)
- `npm run test:mssql` → lines similar to:

- `Connecting with Windows login to Scouting on localhost`

- `OK: connected to Scouting as FDB\Scouting`

- A SQL Server version line

If you see that output, the database layer is healthy. Restart the app only if the website still errors after a config change (see §7).

5.2 If the connectivity test fails — reconnect checklist

Work through the steps below in order. Stop when `npm run test:mssql` succeeds, then restart the Node app (§7) and retry login on https://scouting.kongoniriverfarm.com/

Step A — Confirm Windows identity

```
whoami
```

Result	Action
Shows `fdb\scouting`	Continue to Step B
Shows another user (e.g. `fdb\administrator`)	Sign out and sign in as **`FDB\Scouting`**, or open **Command Prompt → Run as different user** as `FDB\Scouting`, then repeat §5.1

The app and this test **must** use the same Windows account that has rights on the **`Scouting`** database.

Step B — Confirm `.env.local` (production MSSQL settings)

File must exist:

`C:\Users\Scouting\Veg-pro-Smart-Farms-master\.env.local`

Required keys for FDB (values must match the live server):

```
DATA_BACKEND=mssql
NEXT_PUBLIC_DATA_BACKEND=mssql
MSSQL_SERVER=localhost
MSSQL_DATABASE=Scouting
MSSQL_AUTH=windows
MSSQL_TRUST_SERVER_CERTIFICATE=true
SESSION_SECRET=<long-secret>
SESSION_COOKIE_SECURE=true
EVIDENCE_STORAGE_PATH=C:\Users\Scouting\Veg-pro-Smart-Farms-master\data\evidence
```

Quick check (does not print secrets):

```
cd /d C:\Users\Scouting\Veg-pro-Smart-Farms-master
if not exist .env.local echo MISSING .env.local
if not exist .env.local exit /b 1
findstr /B /C:"DATA_BACKEND=" .env.local
findstr /B /C:"MSSQL_SERVER=" .env.local
findstr /B /C:"MSSQL_DATABASE=" .env.local
findstr /B /C:"MSSQL_AUTH=" .env.local
```

Problem	Fix
`Set MSSQL_SERVER in .env.local`	Create or edit `.env.local`; set `MSSQL_SERVER=localhost` and keys above
`MSSQL_AUTH=windows` missing	Add `MSSQL_AUTH=windows` (production on FDB does **not** use SQL password login for the app)
`MSSQL_SERVER=192.168.16.14` on FDB	Change to **`MSSQL_SERVER=localhost`** for Windows Integrated auth on the server (LAN IP often causes "untrusted domain" errors)
Supabase URL keys present	Remove or comment out `NEXT_PUBLIC_SUPABASE_*` and `SUPABASE_SERVICE_ROLE_KEY` on production

After any `.env.local` change, restart the app (§7).

Step C — Confirm SQL Server is running

On FDB (IT / admin):

```
sc query MSSQLSERVER
```

Result	Action
`STATE : 4 RUNNING`	Continue to Step D
Stopped / not found	Start the **SQL Server (MSSQLSERVER)** service in **services.msc**, or ask VegPro IT to start SQL Server on FDB

If your instance is named (not default), use `sc query MSSQL$` instead.

Step D — Test SQL outside Node (Windows auth)

Still as **`FDB\Scouting`**:

```
sqlcmd -S localhost -E -d Scouting -Q "SELECT DB_NAME(), SYSTEM_USER"
```

Result	Action
One row: `Scouting`, `FDB\Scouting`	SQL Server and permissions are OK → go to Step E
`sqlcmd` not recognized	Install **SQL Server Command Line Tools** or use **SSMS** → **New Query** with the same query while logged in as Scouting
Login failed for user	IT must grant **`FDB\Scouting`** access to database **`Scouting`** (db_datareader/db_datawriter or role as deployed)
Cannot open database "Scouting"	Database missing or offline → Step F

Optional: open **SQL Server Management Studio (SSMS)** → connect to `localhost` → Windows Authentication → confirm database `Scouting` exists and is **Online**.

Step E — Node driver and app folder

From the app folder as `FDB\Scouting`:

```
cd /d C:\Users\Scouting\Veg-pro-Smart-Farms-master
node -v
npm run test:mssql
```

Error / symptom	Likely cause	Fix
`Cannot find module 'mssql'` or `msnodesqlv8`	Dependencies not installed	`npm install` (on FDB, use the same install method used at deploy time)
ODBC / driver errors	Missing ODBC driver	Install **Microsoft ODBC Driver 17 for SQL Server** (or 18) on FDB
`Login failed` / `untrusted domain`	Wrong server name or wrong Windows user	`MSSQL_SERVER=localhost`, run as `FDB\Scouting` (Steps A–B)
Timeout / ECONNREFUSED	SQL not listening or wrong port	Step C; confirm port **1433** locally (do **not** expose 1433 to the internet)
Works in `sqlcmd` but fails in Node	Stale env or wrong working directory	Confirm `.env.local` in app root; restart command prompt; `npm run test:mssql` again

Step F — Database missing or empty schema

If `Scouting` does not exist or tables are missing, apply repo scripts **in order** (SSMS → database `Scouting`, or from FDB app folder when VPN/network allows):

```
cd /d C:\Users\Scouting\Veg-pro-Smart-Farms-master
npm run db:mssql:apply
npm run seed:mssql-users
```

Manual equivalent: run each file under `sql/mssql/` in numeric order (`001` … `007`) in SSMS.

Then repeat:

```
npm run test:mssql
```

Step G — After SQL connects again — restart the website app

```
schtasks /End /TN "VegPro App"
schtasks /Run /TN "VegPro App"
netstat -ano | findstr :3000
type C:\ProgramData\VegPro\vegpro-app.log
```

Smoke-test: https://scouting.kongoniriverfarm.com/auth/login

Step H — Testing from a laptop (not on FDB)

Location	What to use
On FDB as `FDB\Scouting`	§5.1 — authoritative production test
Developer PC on VegPro VPN	May use `MSSQL_SERVER=192.168.16.14` with **SQL login** in `.env.local` only if IT provided `MSSQL_USER` / `MSSQL_PASSWORD`; run `npm run test:mssql` from the repo. This is for setup/troubleshooting, not the IIS production `.env.local` on FDB
Off VPN / internet only	SQL to FDB will fail by design (port 1433 is not public). Use the public website URL only

Escalation (when still failing)

Collect and send to VegPro IT / support (no passwords in email):

1. Output of `whoami`
2. Output of `npm run test:mssql` (full error text)
3. Output of `sqlcmd` test (Step D) or SSMS screenshot
4. Last 30 lines of `C:\ProgramData\VegPro\vegpro-app.log`
5. Confirmation whether `.env.local` has `MSSQL_SERVER=localhost` and `MSSQL_AUTH=windows`

5.3 Master data changes

Farms, greenhouses, and varieties are **not** edited from Admin UI in this release. Changes are applied in SQL (`dbo.farms`, `dbo.greenhouses`, `dbo.crop_varieties`, etc.) and may also require updating app master lists and rebuilding. Follow a controlled change process.

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6. IIS / HTTPS checklist

- Site **VegPro** Started
- Bindings: host header `scouting.kongoniriverfarm.com` on **80** and **443**
- SSL certificate: company wildcard `*.kongoniriverfarm.com`
- ARR proxy enabled; `preserveHostHeader=true`
- `web.config` rewrites to `http://127.0.0.1:3000/{R:1}` and sets forwarded proto/host variables
- Windows Firewall allows inbound **80** and **443**
- Do **not** publish SQL port **1433** to the public internet

6.1 Next.js behind IIS

`next.config.ts` includes:

```
experimental: {
  serverActions: {
    allowedOrigins: ["scouting.kongoniriverfarm.com", "127.0.0.1:3000"],
  },
},
```

This is required so login and other Server Actions work through the reverse proxy.

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7. Application auto-start

Scheduled Task `VegPro App`:

- Trigger: At system startup (short delay)
- Run as: `FDB\Scouting`
- Action: `cmd.exe /c C:\ProgramData\VegPro\start-vegpro.cmd`
- Starts: `node .\node_modules\next\dist\bin\next start`

Manual restart

```
schtasks /End /TN "VegPro App"
schtasks /Run /TN "VegPro App"
```

Check app is up

```
netstat -ano | findstr :3000
type C:\ProgramData\VegPro\vegpro-app.log
```

IIS starts with Windows automatically. The Node app depends on this scheduled task.

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8. Deploy / update procedure

1. Put updated code in `C:\Users\Scouting\Veg-pro-Smart-Farms-master` (keep `.env.local`).
2. In that folder:

```
npm install
npm run build
```

1. Restart the app task:

```
schtasks /End /TN "VegPro App"
schtasks /Run /TN "VegPro App"
```

1. Smoke-test: open https://scouting.kongoniriverfarm.com/auth/login and sign in.

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9. Network / DNS behaviour

Client location	How the name resolves	Expected
Public internet / mobile data	Public IP (e.g. `41.220.120.234`) with 80/443 forwarded to FDB	Site loads
Farm LAN	Should resolve to `192.168.16.14` (internal/split DNS)	Site loads without hairpin issues
If LAN still uses public IP only	Hairpin NAT may time out	Ask IT for internal A record, or temporary hosts file: `192.168.16.14 scouting.kongoniriverfarm.com`

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10. Operational smoke tests (handover)

#	Test	Pass criteria
1	Open live URL on mobile data	Login page loads with valid certificate
2	Admin login	Dashboard opens
3	Create staff (Admin → Staff)	New worker can sign in
4	Worker Field work	Can start round and save a scout stop
5	Manager Scouting	New stop appears in records / pressure views
6	Reboot FDB	After ~1 minute, site works again without manual `npm start`
7	`npm run test:mssql`	Connects as `FDB\Scouting`

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11. Known limitations (honest handover)

1. **Master data UI** — farms / greenhouses / varieties are not maintained from Admin screens; SQL (+ possible app list update) is required.
2. **Approvals** — Admin Approvals UI is present; farm manager primary tools are reports (Dashboard / Scouting / Map / History) and Log spray.
3. **Phone hardware testing** — VegPro noted phones are still being procured; final project closure waits for on-farm device testing with no open bugs.
4. **Demo passwords** must be rotated after training.

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12. Support contacts (fill as needed)

Area	Owner
App / training	Kalpit Patel
FDB / IIS / DNS / certificate	VegPro IT (Imran)
System Admin user	Dev
Farm operations testing	VegPro farm managers + scouts

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13. Document set delivered with this handover

1. **Admin User Manual** — staff, reports, approvals, master-data change process
2. **Farm Manager & Field User Manual** — scout and manager step-by-step procedures
3. **Technical Handover** — this document

Live site: https://scouting.kongoniriverfarm.com/